Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Faria Dairy Inc. #3

Physical address of dairy:

Number and Street	City	County	Zip Code
11495 S Van Allen Road	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X15X-011X-XXXX X205-X15X-013X-XXXX

B. OPERATORS

Faria, Dairy Inc.

Operator name: Faria, Dairy Inc.

Telephone no.: Landline: ; Cellular: (209) 204-2342

Mailing Address Number and Street	City	State	Zip Code
13182 S Robinson RD	Escalon	CA	95320

C. OWNERS

Faria, Dairy Inc.

Legal owner name: Faria, Dairy Inc.

Telephone no.: Landline: ; Cellular: (209) 204-2342

Mailing Address Number and Street	City	State	Zip Code
13182 S Robinson RD	Escalon	CA	95320

Faria, Frank

Legal owner name: Faria, Frank

Telephone no.: Landline: ; Cellular: (209) 204-2342

Mailing Address Number and Street	City	State	Zip Code
13182 S Robinson RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1	0	0	0	0	0
Maximum number	1	0	0	0	0	0
Average number	1	0	0	0	0	0
Avg live weight (lbs)	1	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 17.50 *tons per reporting period*

Total nitrogen from manure: 229.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 160.65 *lbs per reporting period*

Total phosphorus from manure: 37.31 *lbs per reporting period*

Total potassium from manure: 146.18 *lbs per reporting period*

Total salt from manure: 470.85 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,268,000 *gallons*

Total nitrogen generated: 7,900.04 *lbs*

Total phosphorus generated: 2,443.94 *lbs*

Total potassium generated: 40,174.79 *lbs*

Total salt generated: 180,116.56 *lbs*

	5,268,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,268,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Deep Well 13	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
MP 1	36	36	2	process wastewater	X205-X15X-044X-XXXX
MP 2	18	18	2	process wastewater	X205-X15X-044X-XXXX
MP 3	26	26	2	process wastewater	X205-X15X-044X-0000
Totals for areas that were used for application	80	80	6		
Totals for areas that were not used for application					
Land application area totals	80	80	6		

B. CROPS AND HARVESTS

MP 1

Field name: MP 1

10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	396.00 *ton*	Dry-weight		56.4	13,300.00	2,100.00	20,000.00		8.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.00	127.57	20.14	191.84	782.71

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	675.00 *ton*	Dry-weight		67.2	12,500.00	2,050.00	10,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	18.75	153.75	25.22	126.69	799.50

MP 2

Field name: MP 2

10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	198.00 *ton*	Dry-weight		54.8	9,160.00	2,010.00	18,500.00		8.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.00	91.09	19.99	183.96	842.26

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	337.50 *ton*	Dry-weight		61.8	13,200.00	1,900.00	9,940.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	18.75	189.09	27.22	142.39	902.48

MP 3

Field name: MP 3

10/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 10/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	286.00 *ton*	Dry-weight		53.2	12,500.00	2,200.00	19,300.00		9.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.00	128.70	22.65	198.71	976.06

MP 3

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	487.50 *ton*	Dry-weight		60.0	13,200.00	1,840.00	14,200.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	18.75	198.00	27.60	213.00	1,155.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

MP 1 - 10/13/2022: Oats, silage-soft dough

Field name: MP 1

Crop: Oats, silage-soft dough — Plant date: 10/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	47.78	6.32	74.19	1.17	504,000.00 *gal*
Application event totals		47.78	6.32	74.19	1.17	

MP 1 - 06/23/2023: Corn, silage

Field name: MP 1

Crop: Corn, silage — Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	36.80	19.01	283.90	1,309.08	1,134,000.00 *gal*
Deep Well 13	Ground water	2.72	0.00	0.00	197.14	5,865,264.00 *gal*
Application event totals		39.52	19.01	283.90	1,506.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	13.83	5.21	144.82	950.70	735,000.00 *gal*
Deep Well 13	Ground water	2.72	0.00	0.00	197.14	5,865,264.00 *gal*
Application event totals		16.55	5.21	144.82	1,147.85	

MP 2 - 10/13/2022: Oats, silage-soft dough

MP 2 - 10/13/2022: Oats, silage-soft dough

Field name: MP 2

Crop: Oats, silage-soft dough

Plant date: 10/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	47.78	6.32	74.19	1.17	252,000.00 *gal*
Application event totals		47.78	6.32	74.19	1.17	

MP 2 - 06/24/2023: Corn, silage

Field name: MP 2

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	35.44	18.30	273.38	1,260.60	546,000.00 *gal*
Deep Well 13	Ground water	2.72	0.00	0.00	197.14	2,932,632.00 *gal*
Application event totals		38.16	18.30	273.38	1,457.74	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	13.44	5.06	140.68	923.54	357,000.00 *gal*
Deep Well 13	Ground water	2.72	0.00	0.00	197.14	2,932,632.00 *gal*
Application event totals		16.16	5.06	140.68	1,120.68	

MP 3 - 10/14/2022: Oats, silage-soft dough

Field name: MP 3

Crop: Oats, silage-soft dough

Plant date: 10/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

MP 3 - 10/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	49.23	6.51	76.43	1.20	375,000.00 *gal*
Application event totals		49.23	6.51	76.43	1.20	

MP 3 - 06/25/2023: Corn, silage

Field name: MP 3

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	37.75	19.49	291.18	1,342.65	840,000.00 *gal*
Deep Well 13	Ground water	2.72	0.00	0.00	197.14	4,236,024.00 *gal*
Application event totals		40.46	19.49	291.18	1,539.79	

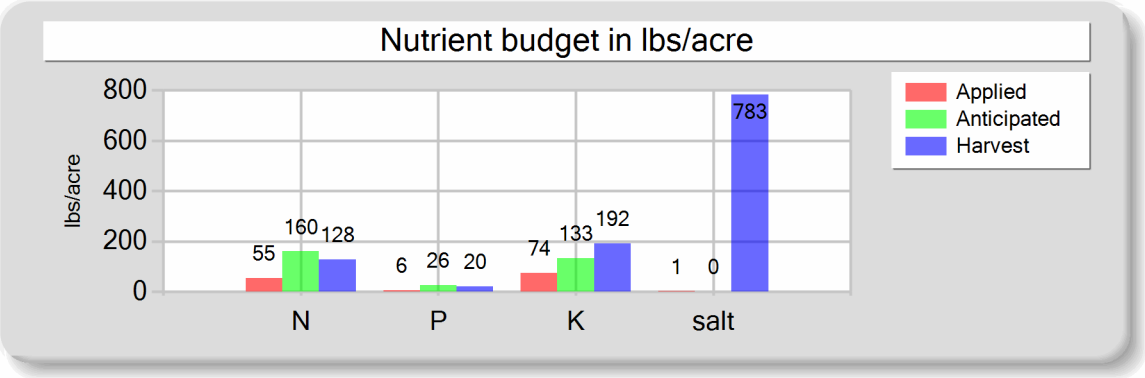
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
wastewater application	Process wastewater	13.68	5.16	143.23	940.26	525,000.00 *gal*
Deep Well 13	Ground water	2.72	0.00	0.00	197.14	4,236,024.00 *gal*
Application event totals		16.40	5.16	143.23	1,137.40	

B. NUTRIENT BUDGET

MP 1 - 10/13/2022: Oats, silage-soft dough

Field name: MP 1 Crop: Oats, silage-soft dough Plant date: 10/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	47.78	6.32	74.19	1.17
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	54.78	6.32	74.19	1.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	127.57	20.14	191.84	782.71
Nutrient balance	-72.79	-13.82	-117.65	-781.54
Applied to removed ratio	0.43	0.31	0.39	0.00

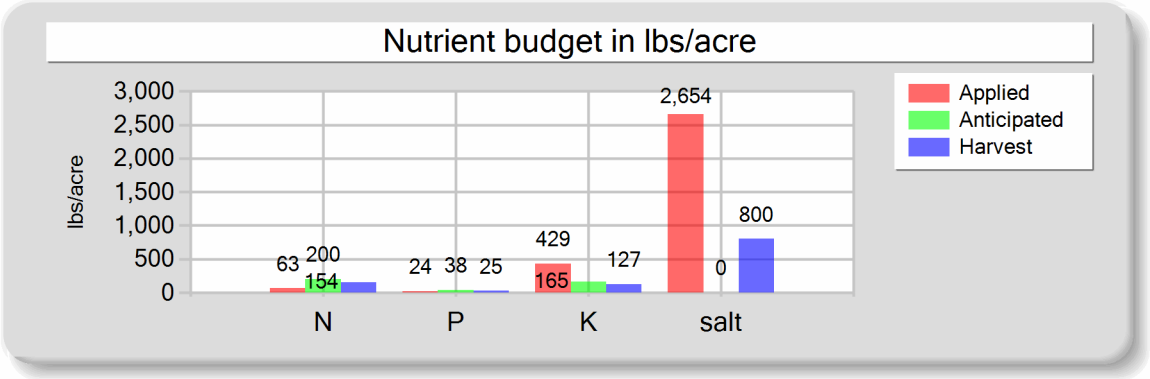
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
504,000.00 *gallons*
18.56 *acre-inches*
0.52 *inches/acre*

Total harvests for the crop
1 *harvests*

MP 1 - 06/23/2023: Corn, silage

Field name: MP 1 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	50.64	24.22	428.72	2,259.78
Fresh water	5.44	0.00	0.00	394.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	63.07	24.22	428.72	2,654.07
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	153.75	25.22	126.69	799.50
Nutrient balance	-90.68	-1.00	302.03	1,854.57
Applied to removed ratio	0.41	0.96	3.38	3.32

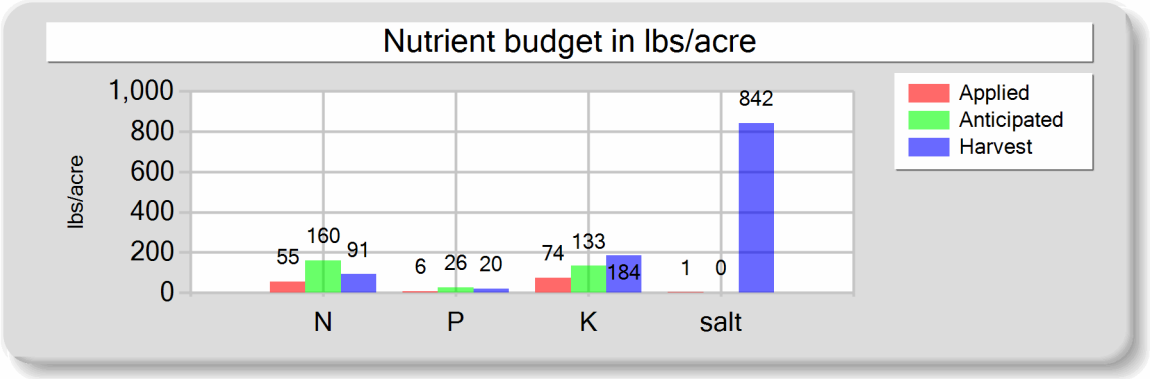
Fresh water applied	
11,730,528.00	*gallons*
432.00	*acre-inches*
12.00	*inches/acre*

Process wastewater applied	
1,869,000.00	*gallons*
68.83	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

MP 2 - 10/13/2022: Oats, silage-soft dough

Field name: MP 2 Crop: Oats, silage-soft dough Plant date: 10/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	47.78	6.32	74.19	1.17
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	54.78	6.32	74.19	1.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	91.09	19.99	183.96	842.26
Nutrient balance	-36.30	-13.67	-109.78	-841.09
Applied to removed ratio	0.60	0.32	0.40	0.00

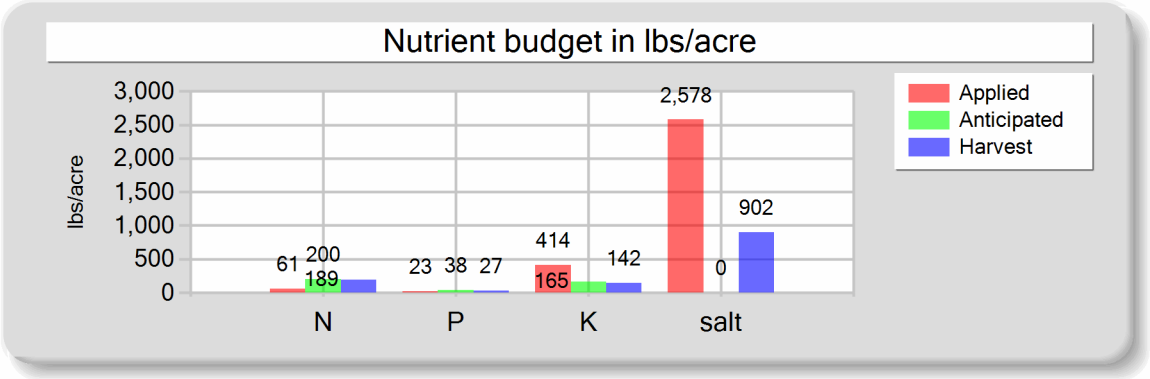
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
252,000.00 *gallons*
9.28 *acre-inches*
0.52 *inches/acre*

Total harvests for the crop
1 *harvests*

MP 2 - 06/24/2023: Corn, silage

Field name: MP 2 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	48.88	23.37	414.06	2,184.14
Fresh water	5.44	0.00	0.00	394.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	61.32	23.37	414.06	2,578.42
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	189.09	27.22	142.39	902.48
Nutrient balance	-127.77	-3.85	271.67	1,675.95
Applied to removed ratio	0.32	0.86	2.91	2.86

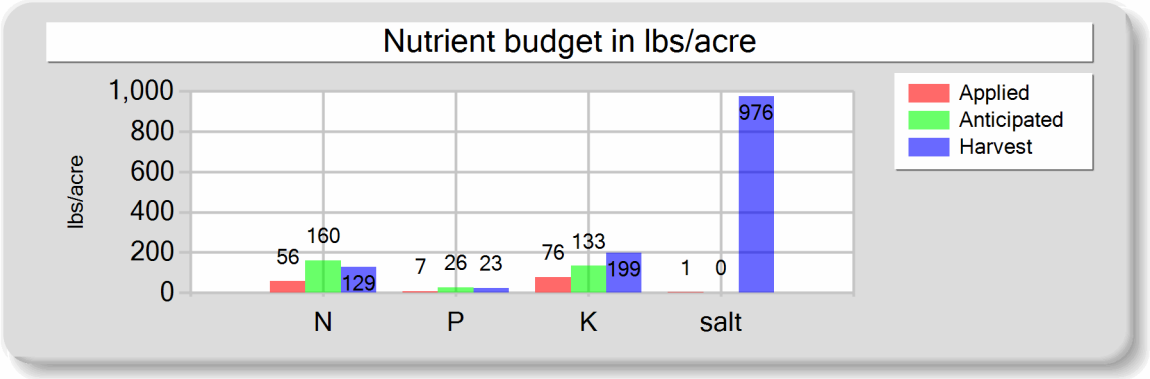
Fresh water applied	
5,865,264.00	*gallons*
216.00	*acre-inches*
12.00	*inches/acre*

Process wastewater applied	
903,000.00	*gallons*
33.25	*acre-inches*
1.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

MP 3 - 10/14/2022: Oats, silage-soft dough

Field name: MP 3 Crop: Oats, silage-soft dough Plant date: 10/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	49.23	6.51	76.43	1.20
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	56.23	6.51	76.43	1.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	128.70	22.65	198.71	976.06
Nutrient balance	-72.47	-16.14	-122.28	-974.86
Applied to removed ratio	0.44	0.29	0.38	0.00

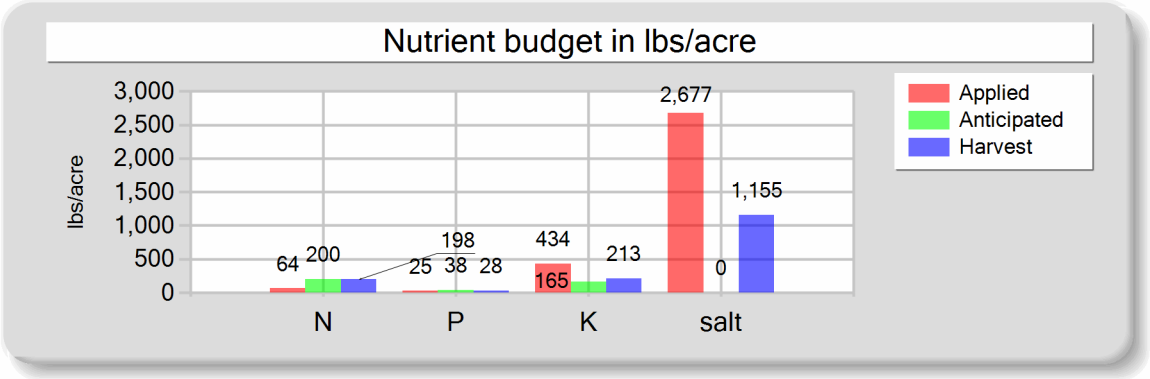
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
375,000.00	*gallons*
13.81	*acre-inches*
0.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

MP 3 - 06/25/2023: Corn, silage

Field name: MP 3 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	51.43	24.65	434.41	2,282.90
Fresh water	5.44	0.00	0.00	394.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	63.87	24.65	434.41	2,677.19
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	198.00	27.60	213.00	1,155.00
Nutrient balance	-134.13	-2.95	221.41	1,522.19
Applied to removed ratio	0.32	0.89	2.04	2.32

Fresh water applied	
8,472,048.00	*gallons*
312.00	*acre-inches*
12.00	*inches/acre*

Process wastewater applied	
1,365,000.00	*gallons*
50.27	*acre-inches*
1.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

t3144027b

Sample and source description: t3144027b

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,400.00	4,110.00	7,360.00	10,600.00	3,840.00	820.00	4,900.00	652.00		18.40
DL	100.00	10.00	10.00	10.00	10.00	10.00	10.00	100.00		0.10

B. PROCESS WASTEWATER ANALYSES

t3053031a

Sample and source description: t3053031a

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	409.00	274.00	0.00	0.00	54.10	635.00								6.13	3,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								0.10	10

t3144027a

Sample and source description: t3144027a

Sample date: 05/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	140.00	44.80	0.00	0.00	72.30	1,080.00								4,690.00	4,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								0.10	10

t3263117a

Sample and source description: t3263117a

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	81.20	11.20	0.00	0.00	30.60	850.00	113.00	106.00	126.00	1,680.00	0.00	160.00	101.00	6,001.00	5,580
DL	0.50	0.50	0.50	0.50	0.20	0.50	0.50	0.50	0.50	2.00	2.00	0.60	0.80	0.10	10

C. FRESH WATER ANALYSES

Deep Well 13

t3174013a

Sample description: t3174013a

Sample date: 06/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.00	0.00	2.00	17.10	10.30	15.10	84.00	0.00	21.60	14.90	271.00	145
DL	1.50	0.50	1.50	0.50	0.50	0.50	2.00	2.00	0.60	0.80	0.10	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

MP 1 - 10/13/2022: Oats, silage-soft dough

MP 1 - 10/13/2022: Oats, silage-soft dough

t3144024a

Sample and source description: t3144024a

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,100.00	20,000.00		8.16
DL	100.00	10.00	10.00		0.10

MP 1 - 06/23/2023: Corn, silage

t3317020a

Sample and source description: t3317020a

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,050.00	10,300.00		6.50
DL	100.00	10.00	10.00		0.10

MP 2 - 10/13/2022: Oats, silage-soft dough

t3144024b

Sample and source description: t3144024b

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,160.00	2,010.00	18,500.00		8.47
DL	100.00	10.00	10.00		0.10

MP 2 - 06/24/2023: Corn, silage

t3317020b

Sample and source description: t3317020b

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	1,900.00	9,940.00		6.30
DL	100.00	10.00	10.00		0.10

MP 3 - 10/14/2022: Oats, silage-soft dough

t3144024c

Sample and source description: t3144024c

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,200.00	19,300.00		9.48
DL	100.00	10.00	10.00		0.10

MP 3 - 06/25/2023: Corn, silage

t3317020c

Sample and source description: t3317020c

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	1,840.00	14,200.00		7.70
DL	100.00	10.00	10.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

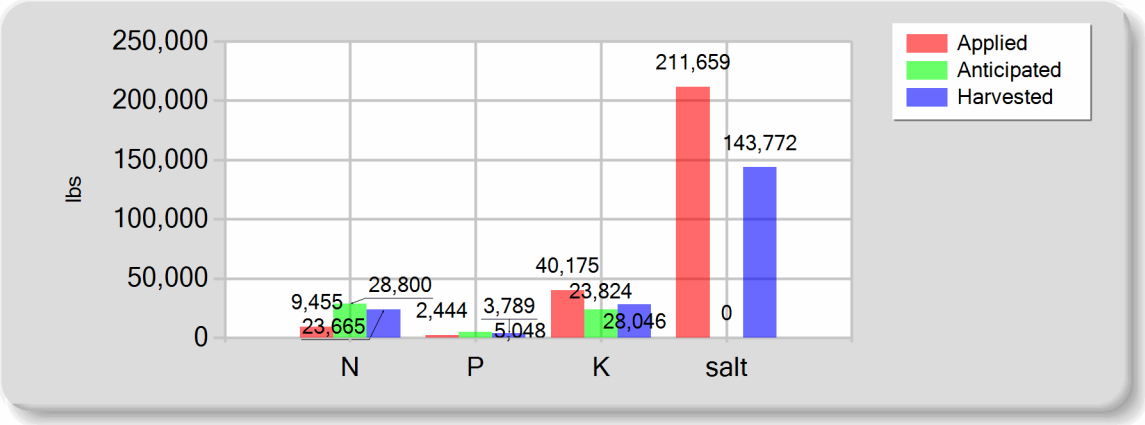
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

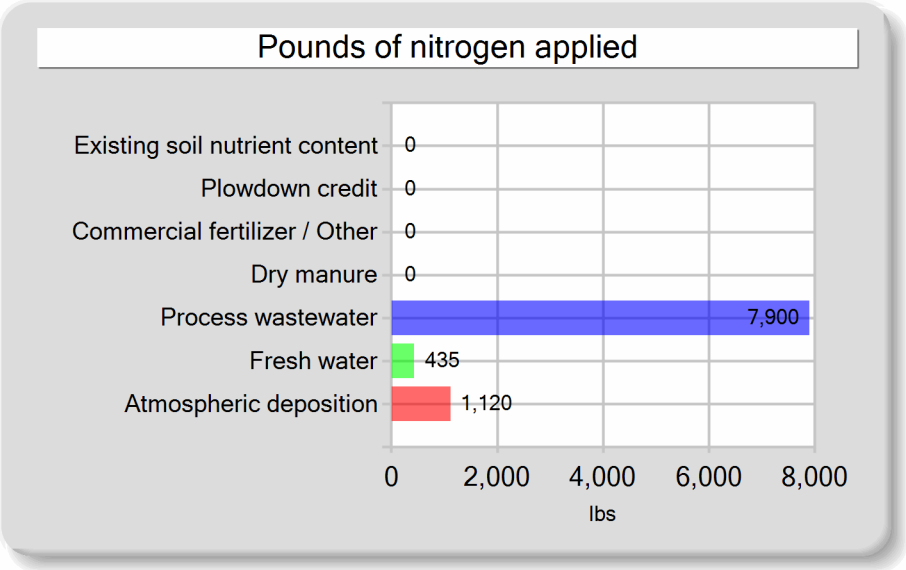
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

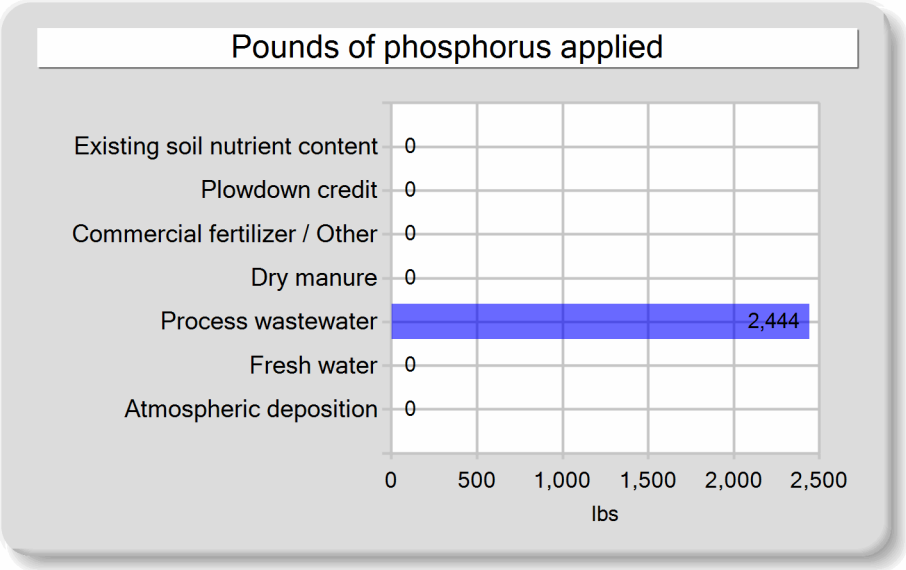
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	7,900.04	2,443.94	40,174.79	180,116.56
Fresh water	435.07	0.00	0.00	31,542.74
Atmospheric deposition	1,120.00	0.00	0.00	0.00
Total nutrients applied	9,455.11	2,443.94	40,174.79	211,659.30
Anticipated crop nutrient removal	28,800.00	5,048.00	23,824.00	0.00
Actual crop nutrient removal	23,665.04	3,789.12	28,045.99	143,772.21
Nutrient balance	-14,209.92	-1,345.17	12,128.80	67,887.09
Applied to removed ratio	0.40	0.64	1.43	1.47

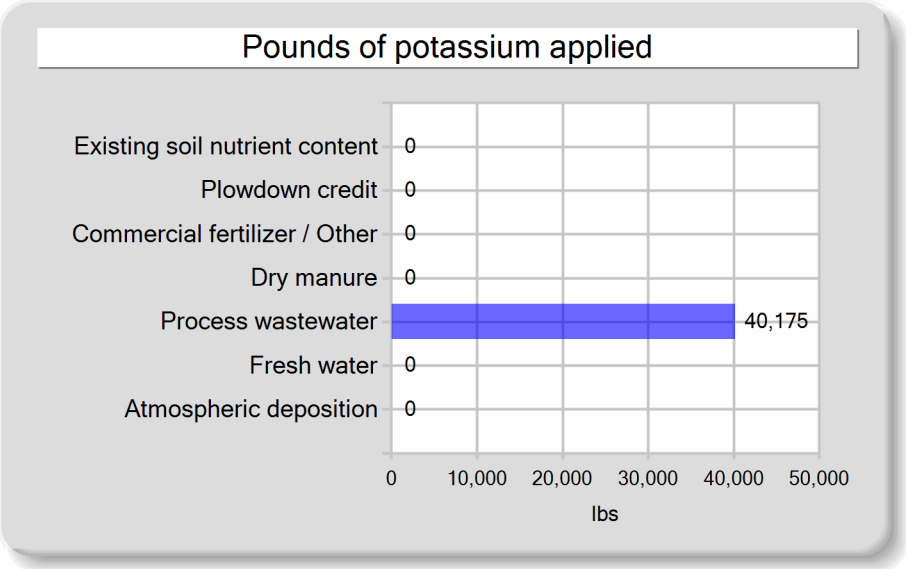
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

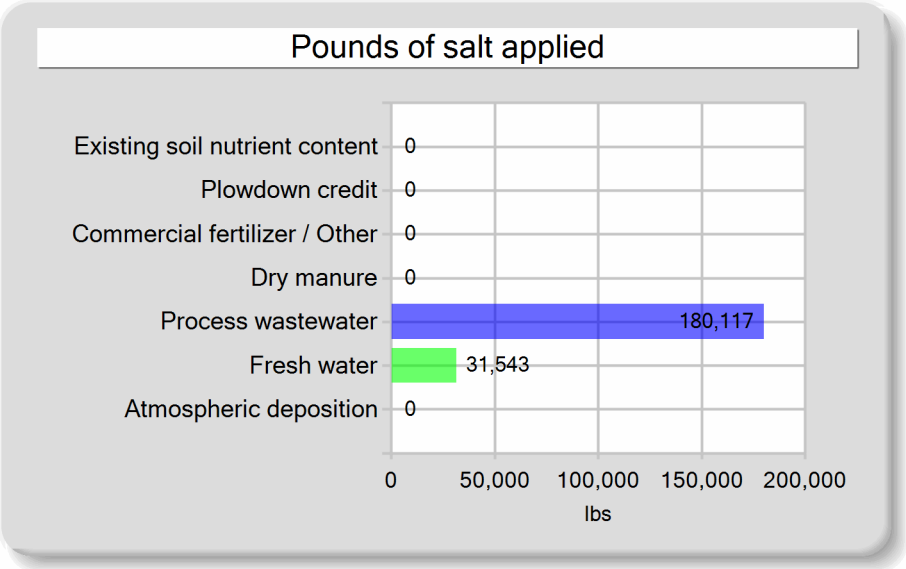


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Dairy was sold 12/14/2023. All cows were removed on 12/31/2022.
Producer had extreme difficulty with irrigation. Only 2 irrigations were completed for corn prior to harvest.
No wastewater was applied in 4th quarter.
Crop yields for Oat Silage and Corn Silage were supported by potential ESN content which contains but is not limited to a combination of possible available soil N from OM and mineralizing organic N.
A 4th quarter wastewater sample was not collected prior to close of escrow.
A second dry manure sample was not collected prior to close of escrow.
Domestic well samples were not collected prior to close of escrow. Samples from 2022 are included for AR submittal purposes. Jakes well is still dry. No sample.
There was no district water in 2023. No sample could be taken.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Faria	Dairy Inc. Faria
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Frank Faria	Frank Faria
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Faria	Dairy Inc. Faria
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-18-24	6-18-24
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
FARIA DAIRY INC
13182 S. Robinson Rd.
Escalon, CA 95320

Grower

Consultant:
CDC

PURCHASE ORDER: #3
RECEIVED DATE: 10/19/22 8:48 AM
SUBMITTED BY: KD
ANALYZED DATE: 10/20/22

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well #14	T2292004A	Electrical Conductivity (EC)	EPA 120.1 and	0.277	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.15	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	177	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.2	106 (T2292004A)	105 (T2292004A)	0.567
Nitrate Nitrogen (NO3-N)	95.4	90.8 (T2293008A)	92.5 (T2293008A)	1.79

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Faria #3

Mailing address:

(not facility address)

Phone#:

Authorized Copy Released To: CDC

Sampled By:

Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER

- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER

- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN

- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER

- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE

- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE

- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL

- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	10/18/22		Domestic well #14				✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	[illegible]	CCS		10/19/22 8:48		
2nd	[illegible]	DENELE ANALYTICAL, INC.	10-19-22 8:48			

(LABORATORY USE ONLY)

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Compliance Analysis Report

Customer
FARIA DAIRY INC
13182 S. Robinson Rd.
Escalon, CA 95320

Grower

Consultant:
CDC

PURCHASE ORDER: #3
RECEIVED DATE: 6/23/23 11:35 AM
SUBMITTED BY: KD
ANALYZED DATE: 6/26/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Deep Well #13	T3174013A	Electrical Conductivity (EC)	EPA 120.1 and	0.271	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	84.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	21.6	mg/L
		Sodium (Na)	EPA 200.7	15.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	10.3	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.00	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	145	mg/L
		Calcium (Ca)	EPA 200.7	17.1	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	14.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	173	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	108			0.000 (T3173009A)	0.000 (null)	
Calcium (Ca)	91.5	120 (T3173018B)	120 (T3173018B)			0.046
Chloride (Cl)	93.4	90.5 (T3174013A)	88.5 (T3174013A)			2.19
Magnesium (Mg)	95.1	124 (T3173018B)	124 (T3173018B)			0.169
Nitrate Nitrogen (NO3-N)	98.2	92.7 (T3174013A)	93.5 (T3174013A)			0.838
Sodium (Na)	108	99.7 (T3173018B)	90.8 (T3173018B)			9.33
Sulfate (SO4)	104	97.7 (T3174013A)	99.8 (T3174013A)			2.15
Total Dissolved Solids (TDS)	96.5			1450 (T3171006A)	1450 (Dup)	0.345

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Faria Dairy Inc #3

Mailing address:
(not facility address)

Phone#:

Authorized Copy Released To: CDC

Sampled By:

Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	6/22/23	9:15	Deep Well #13								H14
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	AD	AI		6/23/23		
2nd	Albert Mata	DENELE ANALYTICAL, INC.	6-23-23 - 11:37			

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)